Reproducibility Engineering
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Chair of Scalable Database Systems
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HDF5 Cheatsheet

This sheet provides a concise overview of HDF5 libraries for Python (h5py). The content is limited to functions that are necessary for solving the tasks of Lab Session 9. You may also use additional functions or HDF5 functions from different libraries (e.g., pandas).

h5py.File(filename, 'w')

Creates a new HDF5 file called `filename`. `'w'` creates a file with write access, overwriting existing files. Other relevant parameters are `'r'` for read-only access and `'r+'` for read/write access for an existing file.

f.create_group(name)

Creates a new group in HDF5 file `f`. Subgroups can be specified by giving the full path, e.g. `'meas/mint/diet'`. Returns a group object.

g.create_dataset(name, data=<data>)

Creates a new data set called `name` in group `g`. The data of the date set is specified in `data`. Pay attention to use an appropriate type, e.g. a numpy array. Returns a data set object.

f[path]

Reads the specified path of HDF5 file `f`. This can return groups, as well as, data sets.

x.attrs[attr] = val

Assigns the value `val` to attribute `attr` in group/data set object `x`.